Automatic Visual Detection of Fresh Poultry Egg Quality Inspection using Image Processing

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***Abstract*— Poultry production is depending on the excellent quality of eggs. Commercially produced eggs are graded concurrently for exterior quality factors like cleanliness and soundness of the shell and inner quality factors like that of the white, yolk, and size of the air cell before being provided to the market in some industrialized countries. However, systems for determining and grading egg quality are still not fully deployed globally at the expected rate. In the Philippines, this method is still challenging for small to medium poultry farms. Owners still practice the traditional method of checking the internal and external quality of eggs which results in time-consuming, requires more manpower, and issues in validation. This paper introduced the design of automatic visual inspection of chicken fresh egg quality both for internal and external quality. Raspberry-pi was employed for a quick and effective output for image processing approach. Different criteria were considered based on the gathered data from the poultry farm owners in developing the system such as the amount of luminance to be used, the safety of egg examination, and accuracy. Different algorithms were employed to perform the internal and external inspections. Based on test results, the camera detection gathered an average of 77.92mm in ten (10) tests before finding the optimal distance for egg detection and detects an average of 75.94 for accuracy. Furthermore, the variance and standard deviation of the data indicate that the values are centered around the mean. Based on the ten (10) sampled tests made, it generated 95% accuracy of correctly detected fresh poultry chicken eggs quality.**

Keywords—image processing, egg quality inspection, egg crack detection, automatic visual detection, raspberry-pi

I. INTRODUCTION

Egg quality has a genetic component, but it also depends on the housing conditions and the hens' housing age. On the farm, eggs that have evident exterior flaws—such as being broken, unclean, or misshapen—are promptly separated. The remaining eggs are candled after this initial examination upon collection to reveal previously missed cracks, weak shells, and interior abnormalities [1]. The productivity of the poultry sector depends heavily on the quality of the eggs. This implies to its various standards that specify the egg's inside and external quality. While the external quality was concentrated on the cleanliness, texture, and shape of the shell, the interior quality focused on the egg white's (albumen) cleanliness and viscosity, air cell size, shape, and strength [2]. Commercially produced eggs are graded concurrently for exterior quality factors like cleanliness and soundness of the shell and inner quality factors like that of the white, yolk, and size of the air cell before being provided to the market in some industrialized countries. However, systems for determining and grading egg quality are still not fully deployed globally at the expected rate [3].

Existing categorization and/or sorting of defective eggs has also been done using a variety of techniques, including optical, mechanical, electrical, and acoustical ones [4]. Popular method today utilized the use of image acquisition to visually detect the internal and external quality of eggs. This technique employs machine vision [4], [5] and deep learning approach [6] [7]. In the Philippines, the method of categorizing and/or sorting in poultry farm are performed manually by poultry farm works for small to medium size farm. This process requires a lot of manpower and prone of damage of egg and error. To validate these issues, two poultry farm owners allowed the researchers to conduct an observations and interview for data collection. The key issue was the difficulty of using the current technology to check the interior and exterior quality of each egg available in the market and high cost. Majority of the research used LED lamp and detecting device pass thru the developed machine with conveyor belt for egg transportation and quality checking using machine vision. These methods were applied to perform egg quality assessment during incubation period [8] [9], execute an evaluation method of eggshell dark spots [10] and grade classification of chicken eggs [11]. These focus on the size and weight of the eggs in assessing and grading using different machine vision algorithms. Finding the optimal distance in egg detection together with the speed of detection and light intensity was not emphasis.

Considering these, the amount of luminance, the inclusion of the part where the internal part of the egg is checked, and the speed of the conveyor, the research aims to accurately detect the internal and external quality of eggs while overcoming the challenges in egg quality detection. In the process of determining the egg's internal and external quality, it also seeks to prevent egg cracking. The researchers

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built an autonomous method for inspecting the quality of poultry eggs to carry out tasks. The system was designed to accommodate chicken eggs also known as table eggs. The device detects the external and internal quality of the eggs one at a time. In developing the system, raspberry-pi was employed. Raspberry-Pi is among the most potent and versatile electrical prototype board that is frequently used today. It works as quickly and effectively as a minicomputer for image processing [12] for different applications [13] [14]. Other electronic components were used such as Arduino that was proven excellent in image and signal processing together with other components such as sensors and LCD monitor [15 -17]. Different criteria were considered based on the gathered data from the poultry farm owners.

In terms of deploying the system for testing and potently used in the market, the system is designed to be mostly beneficial to small and medium sized businesses and local farmers. Multiple eggs are possible but for efficiency purposes when sorting the eggs by defects after, the system needed to process one egg at a time. The reason for this is that typically large scale poultry farms invest on high end machinery to process eggs. Small and medium sized businesses do not see the need for further investments in large machineries if the volume of eggs the producers release is just enough for manual egg candling to still be an efficient method of inspection.

II. Methodology and Experimental Set-up

A. Conceptual Framework

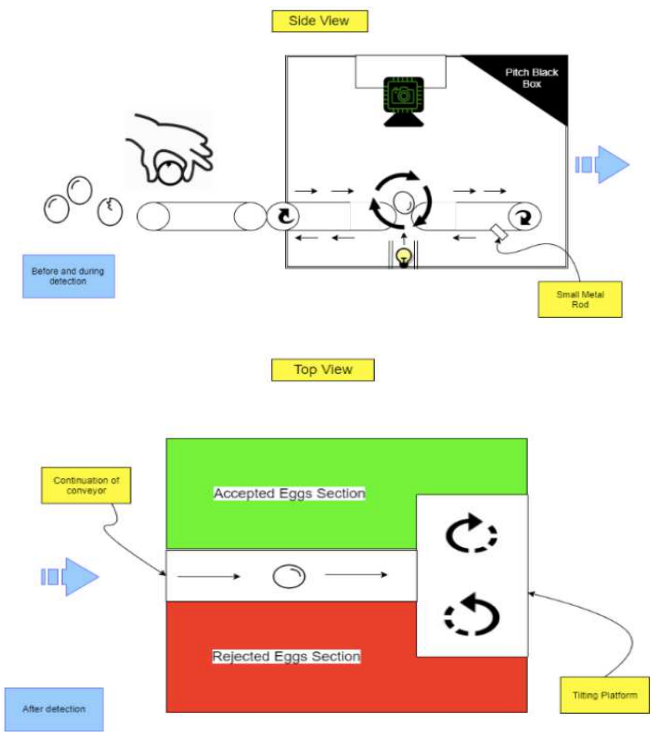

Fig. 1. Conceptual Framework

The chicken egg was manually positioned onto the conveyor. Once an egg is on the conveyor, it will move the egg inside the pitch black box and into a hole that is made by moving the two conveyors just close enough to hold the egg so it won't fall. The egg will be slowly rotated in accordance with the speed of the conveyors in this phase of the design, enabling the camera to fully cover the egg's surface area while it is illuminated and maximizing the detection process's accuracy. It is significant to observe that the second conveyor has a little metal rod attached that is used to push the egg out of the gap between the two conveyors. After the egg has been processed, the second conveyor's small metal rod will snag it from the gap between the two conveyors and transport it onto the tilting platform. The platform tilts both left and right. Depending on the outcomes of the processing, the egg will subsequently be directed on to the appropriate portion (accepted or rejected) by tilting the platform and letting the egg gently glide down the relevant section.

B. System Architecture

The design is divided into four (4) phases: (a) the belt conveyor, which has a servo motor for the stopper and is controlled by the processor; (b) the detection box, which is the heart of the system and has a processor inside to control the camera, led light, LCD touch screen, and all the motors; (c) the chain conveyor; and (d) the paddle, which also uses a DC motor at the conclusion. The egg is tilted and sorted with the paddle.

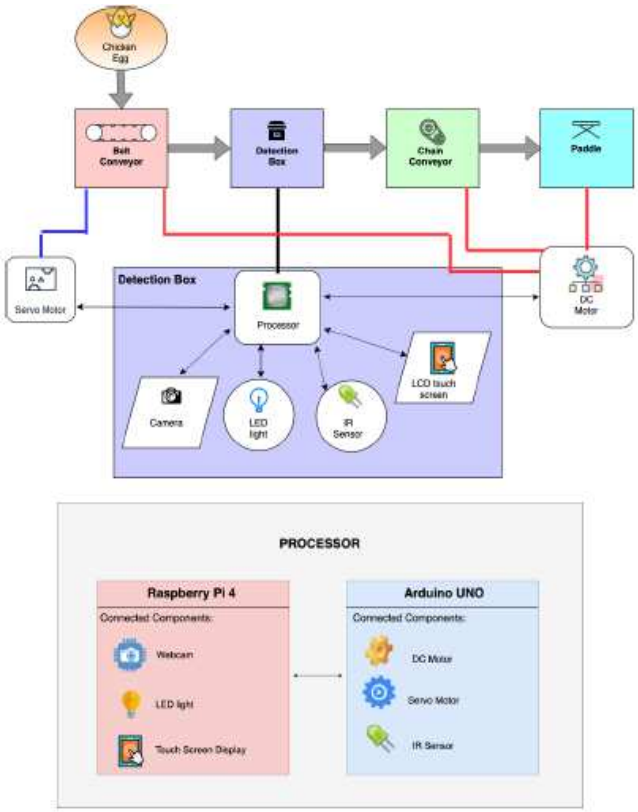

Fig. 2. Design of Automatic Egg Quality Inspection System

The Raspberry Pi serves as the system's primary CPU, and the Arduino Uno serves as the secondary processor. Because the primary detecting program is installed on the Raspberry Pi, it focuses on managing the Webcam input process for the system. The Raspberry Pi is also connected to

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the LCD touch screen, which is used to display and control our system. The system's servo motor and DC motor are among the motors that the Arduino UNO concentrates on controlling. The Arduino UNO software focuses more on each motor's timing, speed, and degree of adjustment. Since the Arduino UNO has the motors connected, it also functions as the Raspberry Pi's receiver. Using serial communication, Raspberry Pi performs the role of the sender in the system. Raspberry Pi delivers a signal to the Arduino UNO with a 1 or a 0, and Arduino UNO then controls the motor in response to the signal it receives.

C. Hardware System Design

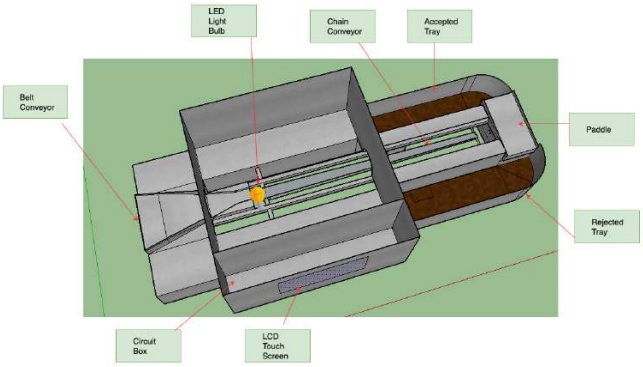

Fig. 3. Top Orthographic projection of the hardware design

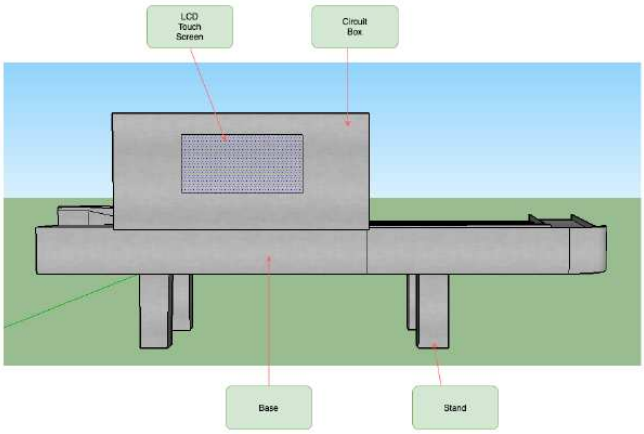

Fig. 4. Orthographic side projection of the hardware design

In the orthographic design, the base, the entire conveyor system, and the detecting box are included. To keep the eggs from becoming slippery and to prevent egg breaking, the belt conveyor at the entrance is made of rubber. The base's midsection is where the detection box is attached, and the circuit box is attached to the side of the detection box. The detection box serves as the primary processing hub for checking the quality of eggs. Aluminum was used in its construction to give it impermeable and odorless qualities. It is non-toxic and frequently used to package delicate goods like food, notably eggs, and medications. The stand was made of steel, and its structures can support a large amount of weight. The circuit box contains every electrical component and has an acrylic plate covering on the front. as seen in Fig. 5 and Fig 6. Illustrate the entrance and exit of the egg.

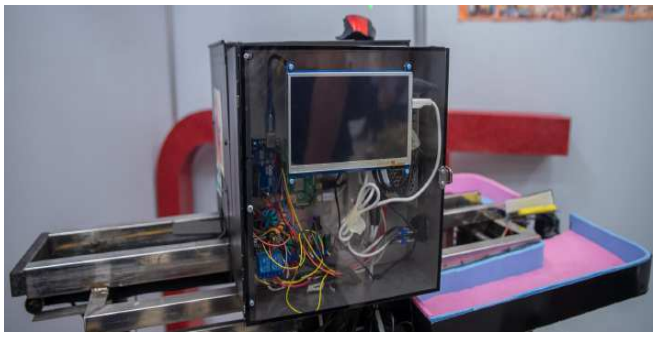

Fig. 5. Actual Hardware Design

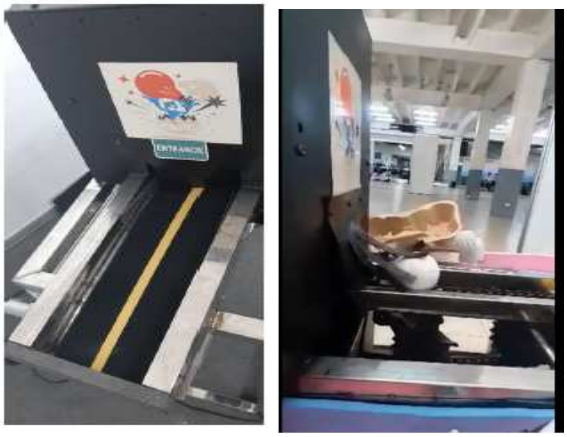

Fig. 6. Placement of egg in the entrance (left) and exit (right) to the conveyor after detection.

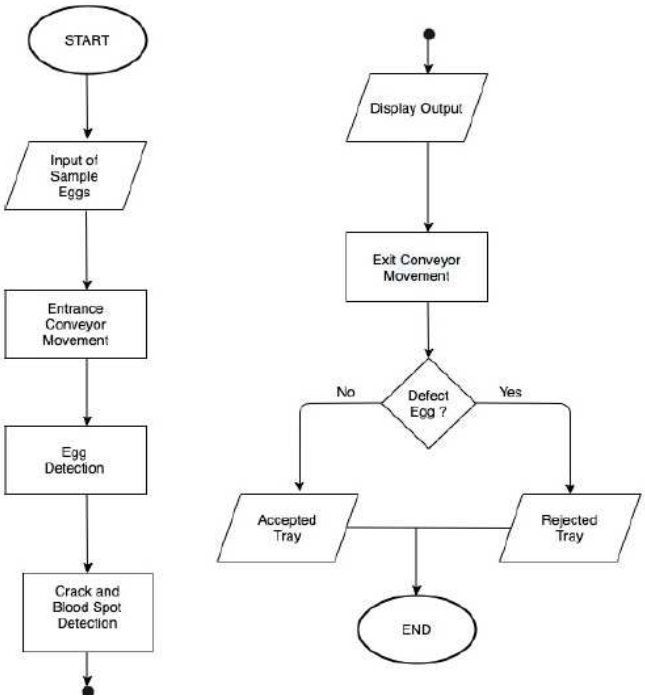

Fig. 7. System Flowchart

Once an egg is detected as a result of the detecting process. A defect or no defect was automatically determined based on the state that was shown on the LCD screen. The system will send a signal to the IR sensor at the end of the conveyor after determining whether or not the egg is

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defective, allowing the paddle to decide whether it will pass through the acceptable tray or not.

D. Hardware Design Criteria

Accuracy of detection, conveyor speed for safety, and speed in terms of distance from the object were considered when the research was completed. Standard Deviation was used to measure the accuracy of external and internal detection. Shown in Table I is the criteria, factors and experimental values used in completing the research.

TABLE I. UTILIZED HARDWARE DESIGN CRITERIA

Criteria	Factors	Experimental Values
Accuracy	Distance	76.2mm Object distance to lens
Safety	Speed of Detection	1920 x 1080 Resolution
Speed	Intensity of Light	60 watts power produced by the bulb
		4 seconds duration of an egg above the light source
	Speed of Intensity of Conveyor	25 m/s
		4 seconds duration of an egg above the light source

Eggs can bear a surprising amount of pressure despite the fact that when external force is applied at a specific point, eggs break easily. Once the chicken farmer has neglected to address cracks, contamination may result from them. The eggs must not crack or cook while being detected, as this could compromise the process. As a result, when addressing egg safety, particularly when it comes to commercial production, the conveyor belt's speed was thought to have a significant impact. The conveyor belt's rapid rotation could cause the ten eggs to crash against one another or the rollers, potentially cracking the conveyor belt.

Speed of detection is a feature that is particularly significant in the farm since it increases the amount of production to meet the demands and satisfy the needs of the consumer. The distance between the capturing device and the object is one aspect that could significantly impact how quickly an object is detected. The ratio between the focal length and distance to the egg is the same as the ratio between the size of the object on the capturing device and the size of the egg.

E. Egg Quality Detection

The egg was checked first for exterior cracks. The crack identification technique makes use of Canny Edge detection. Canny edge detection is a method for obtaining valuable structural information from various visual objects and significantly reducing the amount of data that needs to be processed. The edge detection operator known as Canny employs a multi-stage algorithm to find a variety of edges in photos, most notably the break in the egg. The use of a threshold value of 30,110 was adequate for improving the crack and dirt's visibility.

The obtained raw images of the eggs obtained from the camera were converted from RGB color space to YIQ color space for an optimal result. It has been demonstrated that when the YIQ color space is applied to the functionalities of the system, even the tiniest of shadows present on the surface of the eggs caused by the tiny cracks, breaks, or damages on the egg will be highlighted and revealed. The YIQ color space is designed to take advantage of human color-response characteristics. To further emphasize and let the user observe the eggs, the remaining algorithmic processing steps are added.

The egg was then examined internally. The background of the eggs was blurred using the images that were captured using median blur. Blood spots are identified and improved through the extraction of the red component. The egg yolk, if any, are now plainly identifiable. By using a binary and gaussian filter, blobs are produced that represent the yolks. Additionally, this section displays the results of the interior egg image detection.

F. System Simulation

Sample eggs came from a nearby poultry farm in Silang, Cavite, Philippines, where the research's observation and interview phases were carried out. A total of 30 egg were used. The eggs were physically examined, went through a single multiple inspection for reference in loading them onto the conveyor, and underwent a single test. Fig. 10 illustrates the GUI of the system.

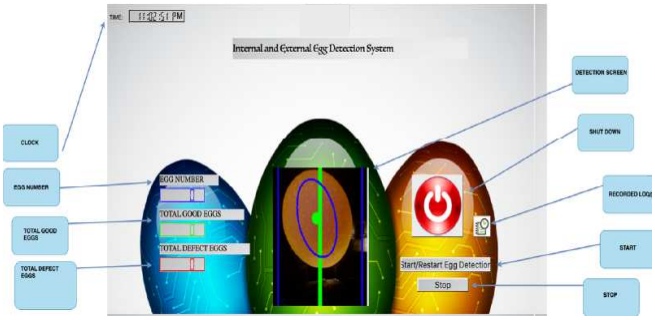

Fig. 8. GUI of the Egg Quality Inspection System

Fig. 9. Sampled Eggs

For the simulation to start, the eggs were prepared. One at a time, the conveyer at the entryway was loaded with eggs.

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Automatically, the LCD panel displays the external and internal quality of the identified egg. The exit conveyor will remove the tested egg following the detection procedure. The system will transmit a signal to the IR sensor at the end of the egg after it has been determined whether it is defective or not, allowing the paddle to decide whether the egg will be accepted or rejected. After completing the process, system's logs contain all information it has collected, including the number of eggs found overall, the number of good and poor eggs, as well as the date and time.

III. Result and Discussion

A total of 10 testing was performed. Shown in Fig. 10 is the output the egg quality detection result. It demonstrates the filtered image to get the blob of the sample egg and the egg detection status. The image was converted into an array and decoded into OpenCV usable format. The two blue contour lines are the detection boundary lane where the process works and the egg counter for entrance and exit conveyor. After getting the blob of the sample egg it draws the oval blue contour and its shows the size area of the sample egg. Fig. 11 exhibits the detected crack line of the fresh poultry egg.

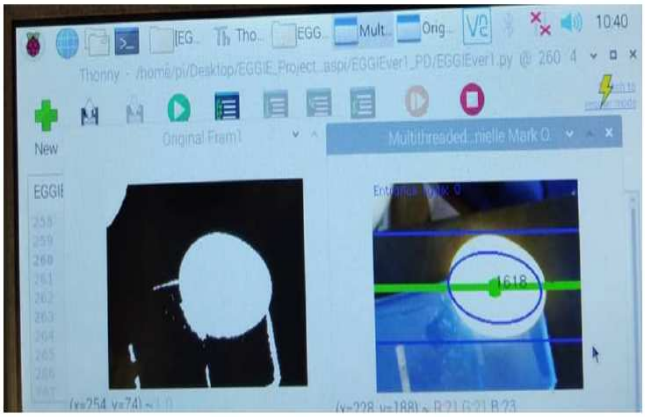

Fig. 10. Result of Egg Detection

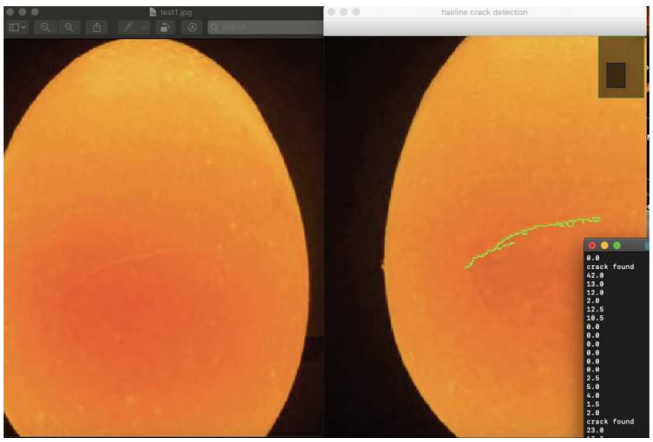

Fig. 11. Hairline of the Detected Egg Crack

The experimental findings demonstrate that, with the employed threshold, improved Canny edge detection can balance removing noise from getting more edge information, which has the well continuity of the edge detection, and can detect the edge detail of the image particularly the cracks inside. The green line is drawn and identified as the crack inside the egg after the algorithm has been applied to the sample eggs.

TABLE II. Statistical Parameters of the Camera Distance

Statistical Parameters	Distance (mm)	Accuracy
Mean	77.922	75.984
Median	74.61	76.325
Mode	76.22	59.34
Range	67	34.26
Variance	330.41	130.17
Standard Variation	18.17	11.41

TABLE III. Statistical Parameters of Image Resolution and Frames per Second

Statistical Parameters	Image Resolution	FPS
Mean	974	19.167
Median	930	18.415
Mode	1080	17.95
Range	1920	33.85
Variance	308484	85.01
Standard Variation	555.41	9.22

TABLE IV. Summary of Test Conducted

Iteration	Number of Eggs	Detected as Good	Detected as Bad	Correctly Detected	Accuracy
1	3	1	2	3	100%
2	4	2	2	4	100%
3	5	5	0	5	100%
4	7	5	2	7	100%
5	10	9	1	9	90%
6	15	10	5	13	86.67%
7	20	10	10	19	95%
8	25	20	5	25	100%
9	26	19	8	25	96.15%
10	30	21	10	29	96.67%

Based on test results, the camera detection gathered an average of 77.92mm in ten trials before finding the optimal distance for egg detection and detects an average of 75.94 for accuracy. Furthermore, the variance and standard deviation of the data indicates that the values are centered around the mean. Based on the ten (10) sample test made, it generated 95% accuracy of correctly detected fresh poultry chicken eggs quality.

IV. Conclusion

The summary of test results proved that the development of the design for automatic visual detection of fresh poultry egg quality inspection using raspberry pi to perform the image processing produced and good result. This output eliminates the manual quality checking system of the eggs and gives opportunity to small to medium poultry farm to innovate the current process being done in checking egg internal and external quality. The distance and image resolution also shows an acceptable result of detection of the egg. Result of the camera detection gathered an average of 77.92mm in ten (10) tests before finding the optimal distance

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for egg detection and detects an average of 75.94 for accuracy. Furthermore, the variance and standard deviation of the data indicate that the values are centered around the mean. These contributed to a 95% accuracy of correctly detection fresh poultry eggs quality with damaging the shell in an acceptable speed of detection.

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